

How is AI wasting water?

Posted by u/stirfriedquinoa • 0 points • 11 comments

I read all over the place that every time you ask ChatGPT or some other AI a question, it wastes tons of water to cool the processors or something (I don't know computers). But doesn't the water just get evaporated and go back into the clouds to return later as rain? How is it wasted?

Comments

u/Visible-Button8316 • 1 points • 2025-11-25 12:47 UTC

Are the data centres running liquid cooling? Why not use oil like some DIY projects that submerge the motherboard and cpu under oil? That way it just sustains it self.

Granted I have no idea about the efficiency between oil and water. I have a water-cooled computer which just passes through the cooling plate and loops. If this can be achieved on a smaller scale, why can't they build the system to recapture the 'steam' for a lack of bigger understanding, and just reuse the water again and again?

u/DingoFlamingoThing • 1 points • 2025-11-25 12:29 UTC

It takes time and energy to get that water back. You can't just pluck it out of the sky.

Even once it rains and that water does come back to earth, it has to be collected and decontaminated.

u/StitchRecovery • 1 points • 2025-11-25 12:27 UTC

It's not that the water disappears forever, but it can't really be reused right away. Data centers use water to cool their servers, and a lot of it turns into steam and goes into the air. Even if it eventually falls as rain, it doesn't come back where it's needed.

Also, these centers are often in places where water is limited, so using that much for cooling puts extra pressure on the local supply. It's less about losing water completely and more about it being unavailable locally when it's needed.

u/jayron32 • 2 points • 2025-11-25 12:17 UTC

They are using up available potable water from their local areas. That water, before AI, was useful for drinking and irrigation and a number of other human-useful needs. Afterwards, it isn't so useful until it comes back as rain, filters through aquifers, etc. etc. That takes time, and in the meantime, it isn't being useful to the humans that live in that area.

u/ExcessivePlumbing • 1 points • 2025-11-25 12:01 UTC

The water is not removed from our planet, in that sense it is not wasted.

But it might be taken from the same reservoir that is used to feed your household, and it's not returned back to the reservoir.

u/THEbassettMAN • 3 points • 2025-11-25 11:51 UTC

Given that no one else has mentioned it, that water that's used in industrial cooling ends up saturated with the materials used in the piping. Because waste water typically isn't that high in those materials, water treatment plants don't measure/remove them before releasing the water, which ends up being toxic to plant and animal life.

u/CherryBlossomThirst • 7 points • 2025-11-25 11:50 UTC

It's not like AI is drinking water, but running massive data centers uses a lot of water for cooling. Technically, most of it evaporates and eventually comes back as rain, but it's not available locally anymore. So in places facing water scarcity, pulling millions of gallons for cooling can strain resources even if it's technically recycled. The 'waste' is really about timing and location, not disappearing water.

u/CHILINVLN • 1 points • 2025-11-25 11:46 UTC

Water never leaves the planet.

u/Forest_Orc • 4 points • 2025-11-25 11:45 UTC

\>But doesn't the water just get evaporated and go back into the clouds to return later as rain? How is it wasted?

Return as rain, but you don't know where/when. There is only so many water available on the ground which is used for agriculture, industry, and human to drink, at a moment, if you pump it faster than it rains, you're in trouble

u/Petwins • 6 points • 2025-11-25 11:45 UTC

Take a glass of water, pour it down the drain, then wait for the glass to refill by itself and then drink it.

Just because it returns to the water cycle does not mean its available or usable for us. It takes effort, money, and time to capture, clean, and transport water for human use. And that is wasted when you waste clean water.

u/Schnutzel • 11 points • 2025-11-25 11:41 UTC

Wasting water mean wasting *available* water. Yes, the water will come back later as rain, but it takes time.